Session 3: Biological Databases

KB8029 — Bioinformatics & Deep Learning

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SciLifeLab

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Log in to the lecture quiz

Log in to the quiz

Scan the code, or go to quiz.bioinfo.se/live

Log in with your university account

Answering counts as being at the lecture: up to 3 bonus points on the exam

Right or wrong is never graded, only whether you answered. Results are shown without names.

quiz.bioinfo.se/live

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Warm-up: what do you know already?

3 quick questions

On the topics of today, one at a time, about 30 seconds each

Answer on your own, and guess if you do not know

It shows me where to start; right or wrong is never graded

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What is an accession number?

A A stable identifier for one database record

B The official name of the gene

C The length of the sequence

D The year the sequence was published

Answer now
Scan the code, or go to
quiz.bioinfo.se/live

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What does it mean that a database entry is curated?

A It was computed automatically from the sequence

B The authors submitted it themselves

C An expert has checked it against the literature

D It has a 3D structure

Answer now
Scan the code, or go to
quiz.bioinfo.se/live

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A PubMed search for A AND B, compared with A alone, gives …

A the same number of articles or fewer

B the same number or more

C exactly half as many

D a number that only depends on B

Answer now
Scan the code, or go to
quiz.bioinfo.se/live

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What the room knows already

Where we start

Which topics most of you know, and which we take slowly today.

Show the results ▸ quiz.bioinfo.se/live/report

Your questions before today

What you asked

The questions you sent in on Canvas before the lecture

The most common ones now; the others in the block where they belong

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In up to three words: what do you want to learn today?

Identifiers

INSDC, RefSeq, accession.version

Who vouches?

UniProt, InterPro, PDB, AlphaFold DB

Searching

interactions, dossier, PubMed, Boolean

Answer now

Scan the code, or go to
quiz.bioinfo.se/live

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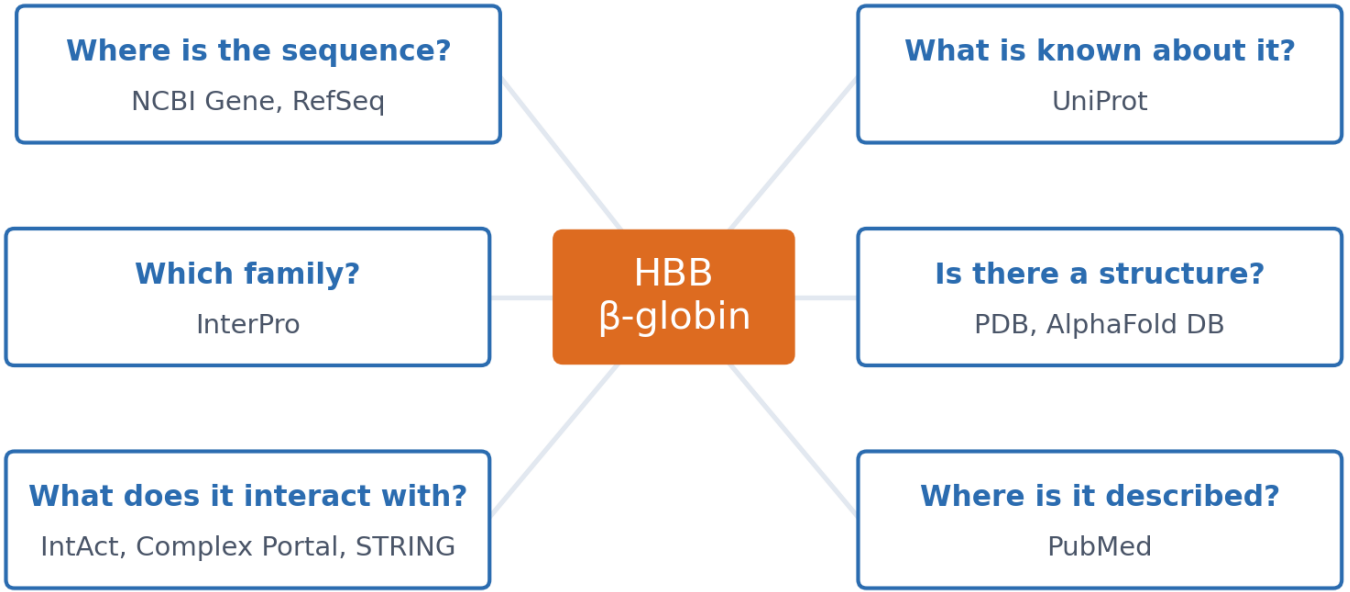
What you want to learn today

What the room wants to learn

Show the results ▸ quiz.bioinfo.se/live/report

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One protein, five questions



We follow human β-globin (HBB) through five questions. Each database appears as the answer to one of them, not as an entry in a list to learn.

See course.bioinfo.se/session03.html, 'Learning goals'.

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Three blocks: about 30 min talk and 15 min discussion



Block 1: shared databases and the identifiers that link them. Block 2: who vouches for a record, and HBB in UniProt, InterPro, the PDB and AlphaFold DB. Block 3: interactions, the dossier, and searching.

Times are approximate: a 10 min start (warm-up quiz and your questions), two 15 min breaks and a 5 min close. Each block ends with a quiz question: guess, run the notebook and work out why it behaves as it does, discuss, answer again.

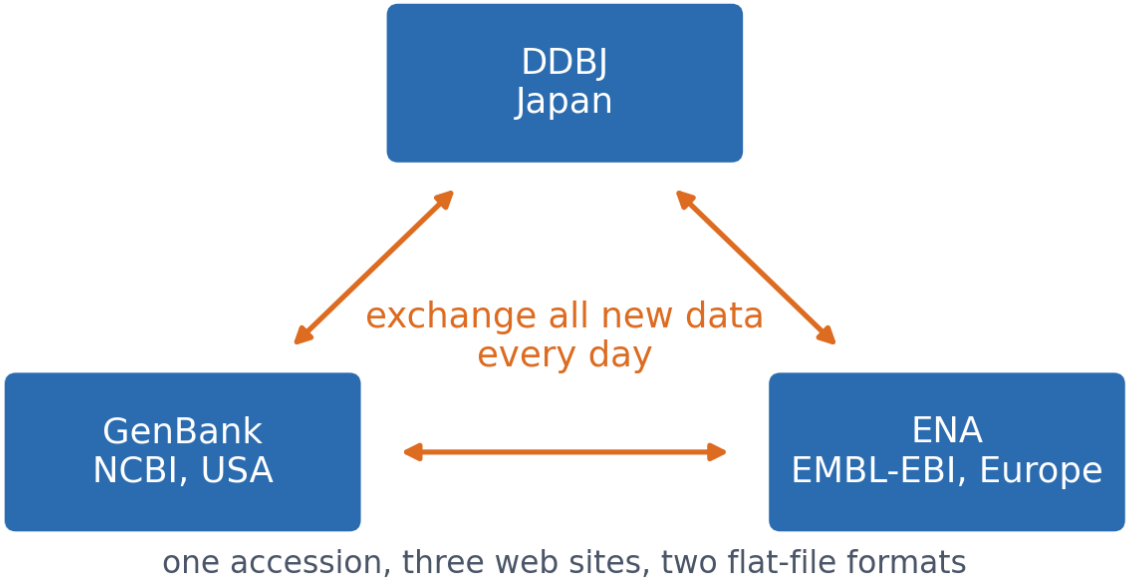
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BLOCK 1 · ABOUT 30 MIN

Shared databases, and the identifiers that link them

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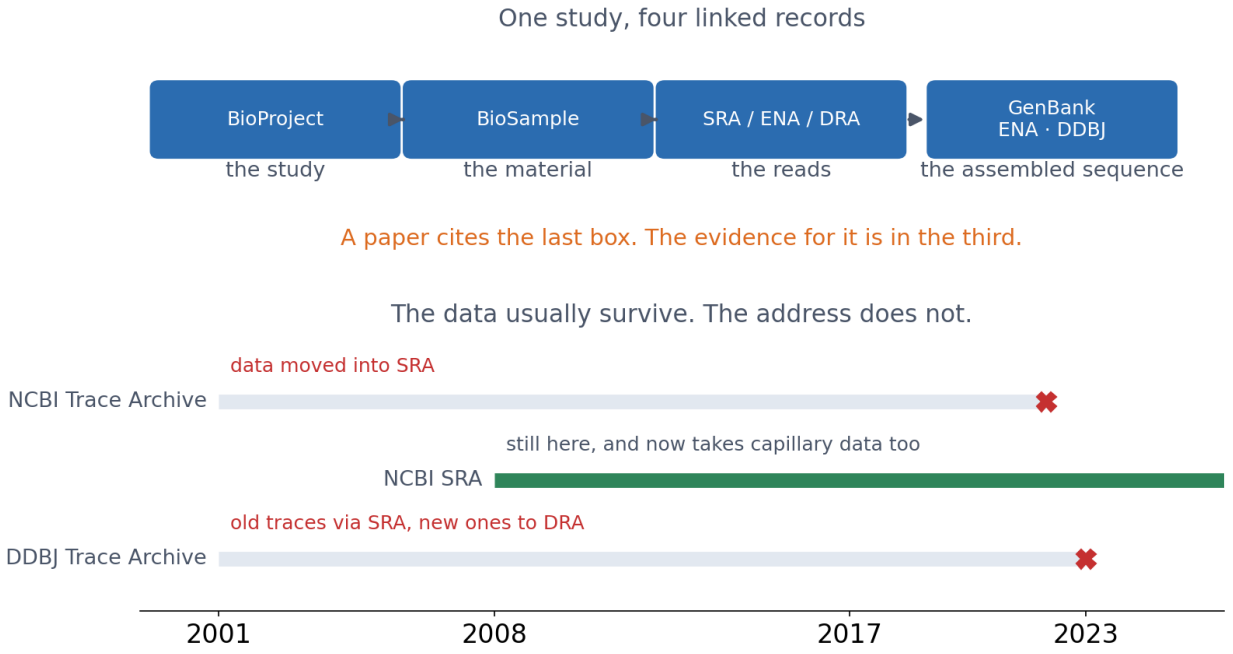
INSDC: one collaboration, three sites



A sequence submitted to any of the three can be found through all three, under the same accession. Access is free; most journals require deposition before publication.

See course.bioinfo.se/session03.html, 'Why coordinated, worldwide databases'.

Behind every sequence: the reads, and where they went

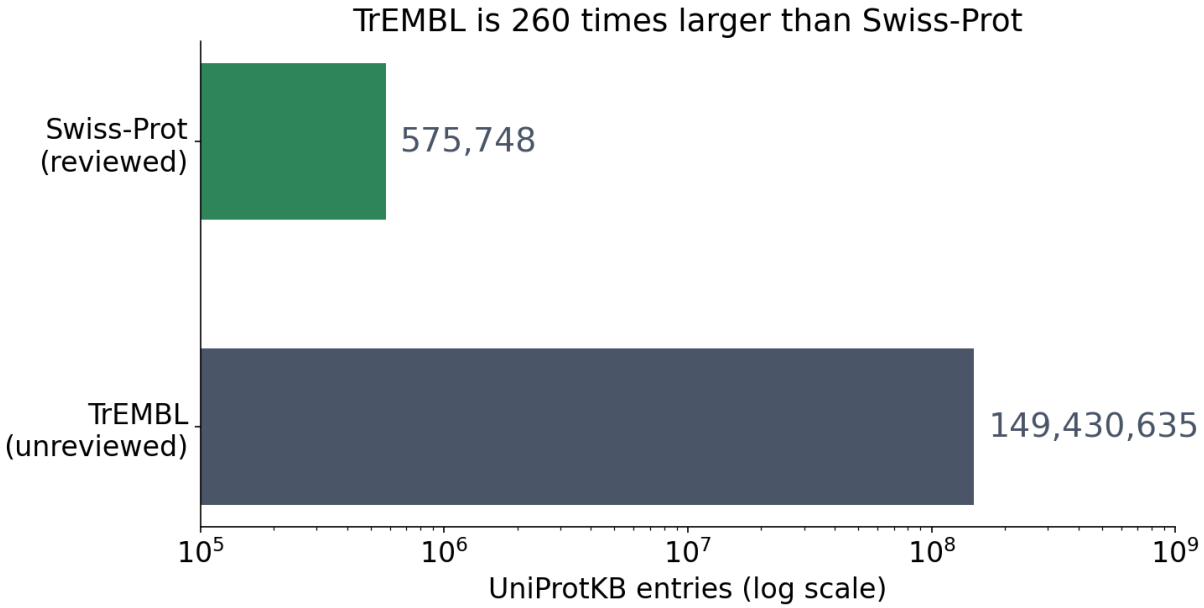


A submitted sequence is the end of a chain. The study, the material and the raw reads are separate records, linked to it — and the archives holding them are not permanent addresses.

NCBI Trace Archive retired 17 June 2022, its data moved into SRA; DDBJ's closed in 2023.

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Too much data for any one lab



- **TrEMBL: ~149 million entries**
- A private copy goes stale at once
- So the databases are shared

Counts fetched live from the UniProt REST API; see the notebook.

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Know which identifier you are holding

HBB	approved gene symbol	**HGNC**
3043	NCBI Gene ID	**NCBI**
NM_000518	transcript record	**RefSeq**
NM_000518.5	one exact sequence (version 5)	**RefSeq**
P68871	UniProt accession (stable)	**UniProt**
HBB_HUMAN	UniProt entry name (can change)	**UniProt**
2HHB	one of hundreds of structures	**PDB**

Seven identifiers for one gene and its products. An accession is unique only within its own database.

See course.bioinfo.se/session03.html, Table 3.1.

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Anatomy of an accession

NM_ 000518 .5

prefix:
mRNA, not predicted

number:
unique within RefSeq

version:
changes with the sequence

report accession.version: NM_000518 alone does not say which revision

NM_000518 is a record; NM_000518.5 is one exact sequence. The version changes whenever the sequence changes.

See course.bioinfo.se/session03.html, 'Identifiers'.

The same identifier in a real record

The mRNA record

```
LOCUS       NM_000518               628 bp    mRNA    linear
DEFINITION  Homo sapiens hemoglobin subunit beta (HBB), mRNA.
ACCESSION   NM_000518
VERSION     NM_000518.5
KEYWORDS    RefSeq; MANE Select.
SOURCE      Homo sapiens (human)
  ORGANISM  Homo sapiens
```

→ the record's name and its length

→ the version: this exact sequence, not just this record

→ MANE Select: the agreed reference transcript

→ which organism the sequence came from

The protein it codes for: a record of its own

```
LOCUS       NP_000509               147 aa            linear
DEFINITION  hemoglobin subunit beta [Homo sapiens].
ACCESSION   NP_000509
VERSION     NP_000509.1
DBSOURCE    REFSEQ: accession NM_000518.5
KEYWORDS    RefSeq; MANE Select.
```

→ a separate accession, with its own version history

→ and the mRNA record it was translated from

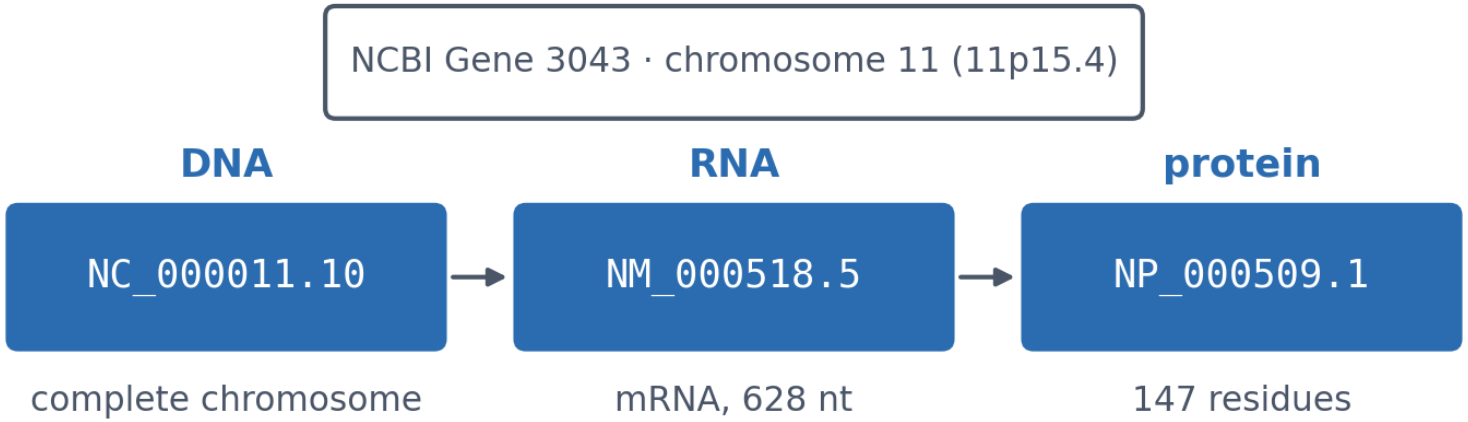
The heads of the GenBank records for NM_000518.5 and NP_000509.1, retrieved 2026-11-06 from NCBI efetch. Public domain (US government work).

The head of NM_000518.5 as NCBI returns it. LOCUS names the record and its length, VERSION fixes the exact sequence, KEYWORDS marks it MANE Select, SOURCE says which organism.

Provenance is printed under the figure itself. The same figure is @fig-s03-record-anatomy in the chapter.

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One gene, three levels, linked by cross-references



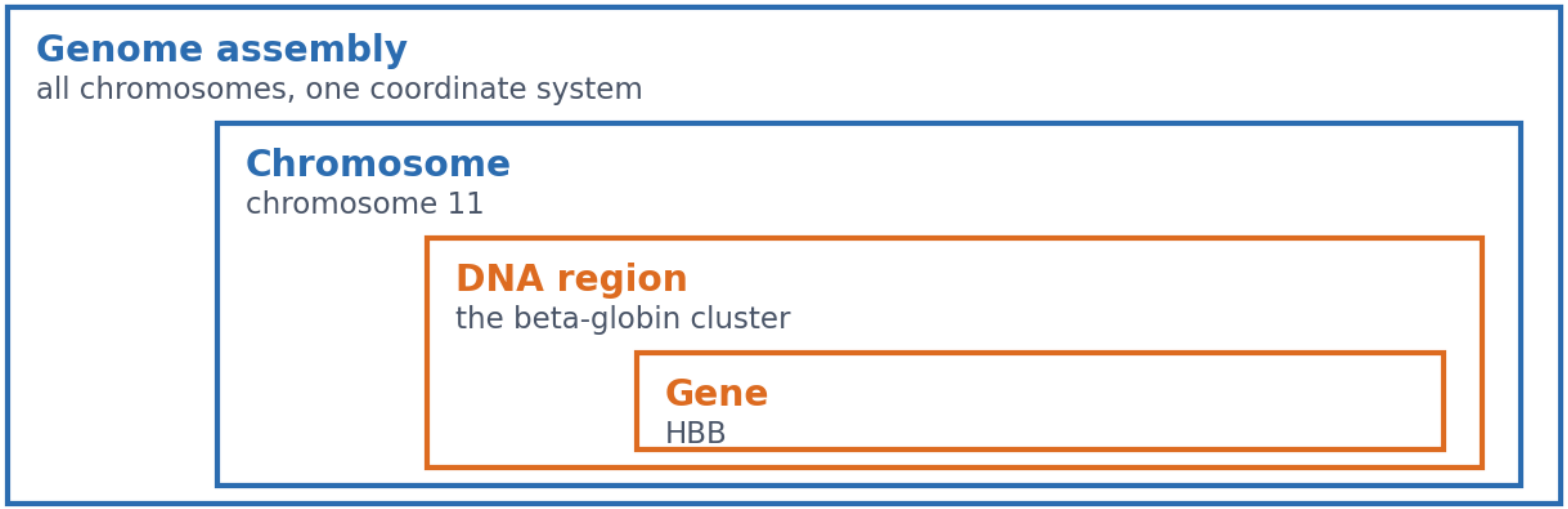
NP_000509.1 says DBSOURCE: REFSEQ: accession NM_000518.5 (a cross-reference)

Databases follow the central dogma: DNA, RNA and protein each have their own records. Genome browsers (UCSC, Ensembl) show the gene in its chromosomal context.

Records fetched live via Entrez; see the notebook.

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Assembly, chromosome, region, gene: one DNA, four zoom levels

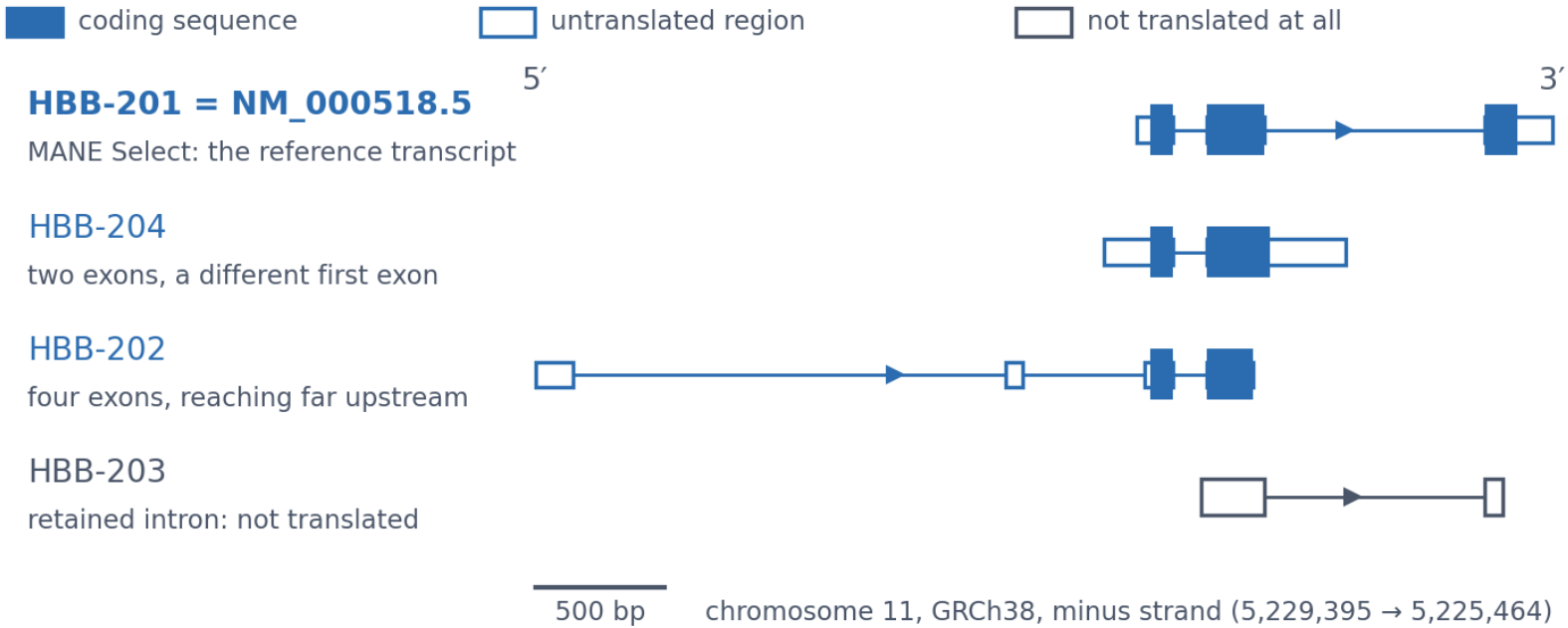


Zooming in changes the region and the detail; the assembly supplies the coordinates throughout.

Not four kinds of thing in a row, but the same DNA at four scales. The assembly supplies the coordinates all the way down.

Drawn by tools/builders/make_session03_book_figures.py; @fig-s03-dna-scales in the chapter.

Four real transcripts of HBB



HBB-201 is the MANE Select transcript, NM_000518.5. The other three differ in their first exon, in how far upstream they reach, and one keeps an intron and is never translated.

Coordinates from Ensembl GRCh38, retrieved 5 November 2026; drawn by tools/builders/make_session03_book_figures.py. @fig-s03-transcript-routes in the chapter.

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RefSeq prefixes: level and kind of record

NC_	genomic: a complete chromosome	NC_000011.10
NM_	mRNA, not predicted	NM_000518.5
NP_	protein, not predicted	NP_000509.1
NR_	non-coding RNA, not predicted	
XM_ / XP_	mRNA / protein PREDICTED	XP_011536997.1

- **The prefix tells the level**
- **X: predicted**
- N: not predicted, many curated

See course.bioinfo.se/session03.html, Table 3.2; NCBI RefSeq FAQ.

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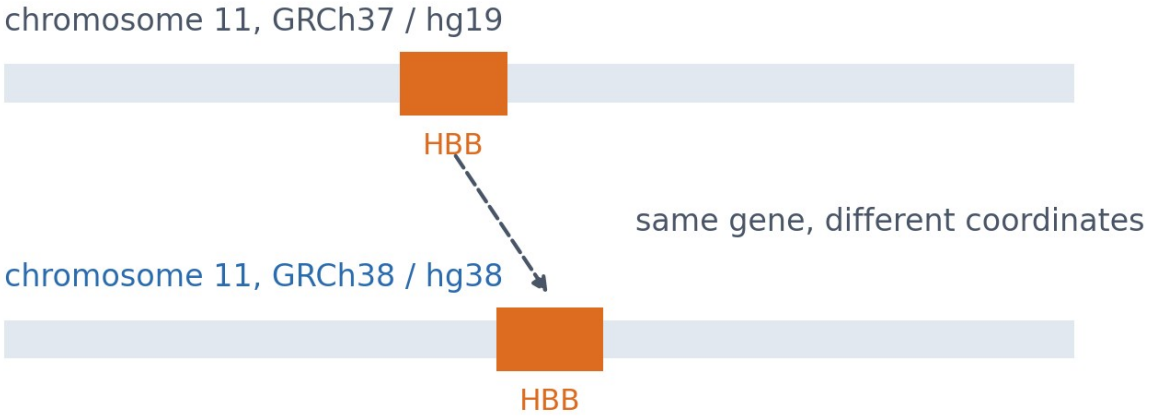
A predicted record says so itself

```
LOCUS       XP_011536997.1
DEFINITION  SWI/SNF-related matrix-associated actin-dependent
            regulator of chromatin subfamily D member 1 isoform X3
COMMENT     MODEL REFSEQ:  This record is predicted by
            automated computational analysis.
```

the record says itself where its information comes from

XP_011536997.1, fetched live in the notebook. Learn to read where a record's information comes from (Block 2).

Coordinates need the genome build

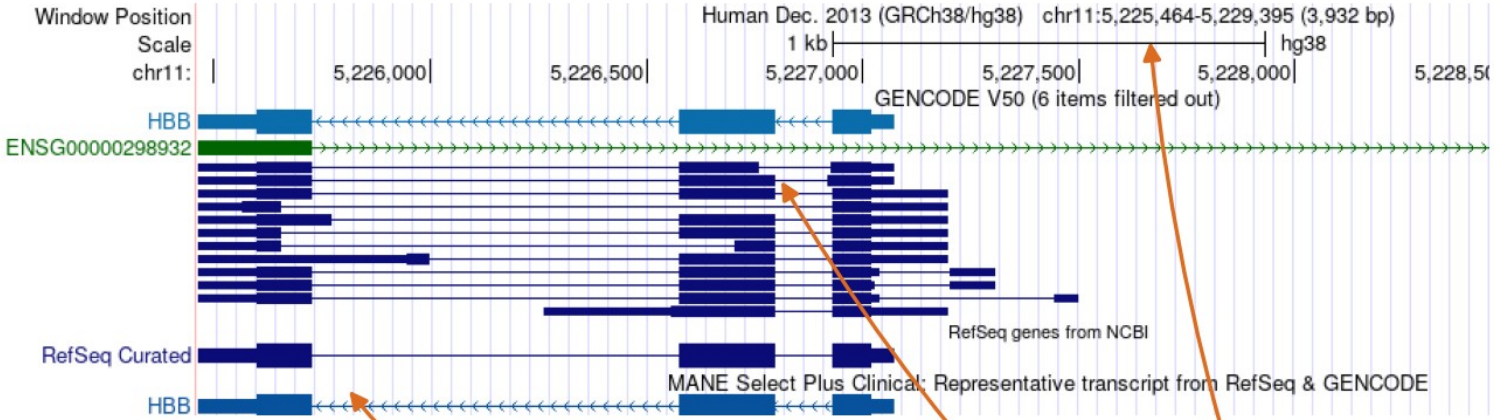


- **GRCh37/hg19 or GRCh38/hg38?**
- Same gene, different positions
- Always report the build

See course.bioinfo.se/session03.html, 'Identifiers'. The lab compares the two builds.

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The same gene in a genome browser



RefSeq curated, and MANE Select = NM_000518.5

GENCODE: every annotated transcript of HBB

the assembly and the exact coordinates

chr11:5,225,464-5,229,395 on GRCh38/hg38. The ruler, every annotated transcript in GENCODE, and the MANE Select row the previous slide named.

UCSC Genome Browser via hgRenderTracks, rendered by tools/builders/make_session03_record_figures.py. The uncropped figure is @fig-s03-ucsc in the chapter.

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Two habits

Moving between records

follow a cross-reference

- Otherwise: stable ID or approved symbol, plus organism
- Then check organism and sequence

Reporting a sequence

accession.version

- NM_000518.5, not NM_000518
- Coordinates with their build

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NM_000546.5 and NM_000546.6: what changed?

A 2015 paper used NM_000546.5 (TP53 mRNA). Today RefSeq has NM_000546.6.

TP53 mRNA in RefSeq

```
2015 paper:   NM_000546.5

today:        NM_000546.6

              ↑          ↑
          accession  .  version
```

What do you know about the two records?

A Same sequence: only the annotation changed

B Same gene and record, but the sequence changed

C Two transcript variants of TP53

D Unrelated: a new version is a new record

Guess on your own

Scan the code, or go to
quiz.bioinfo.se/live

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NM_000546.5 and NM_000546.6: what changed?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

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NM_000546.5 and NM_000546.6: what changed?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session03-discussion-1.ipynb

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NM_000546.5 and NM_000546.6: what changed?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

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NM_000546.5 and NM_000546.6: what changed?

Results, and who explains

Both rounds side by side.

Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 1 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session03-discussion-1.ipynb

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The answer: the same record, a changed sequence



coding sequence: the same 393-residue p53

−60 bases at the 5′ end

−19 bases at the 3′ end

The number after the dot is the sequence version: it goes up only when the sequence changes. Version 6 lost 60 bases at the 5′ end and 19 at the 3′ end; the coding sequence, and so the p53 protein, is the same. A position in the 2015 paper refers to version 5, so always record the full accession.version.

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BLOCK 1 DONE

Break

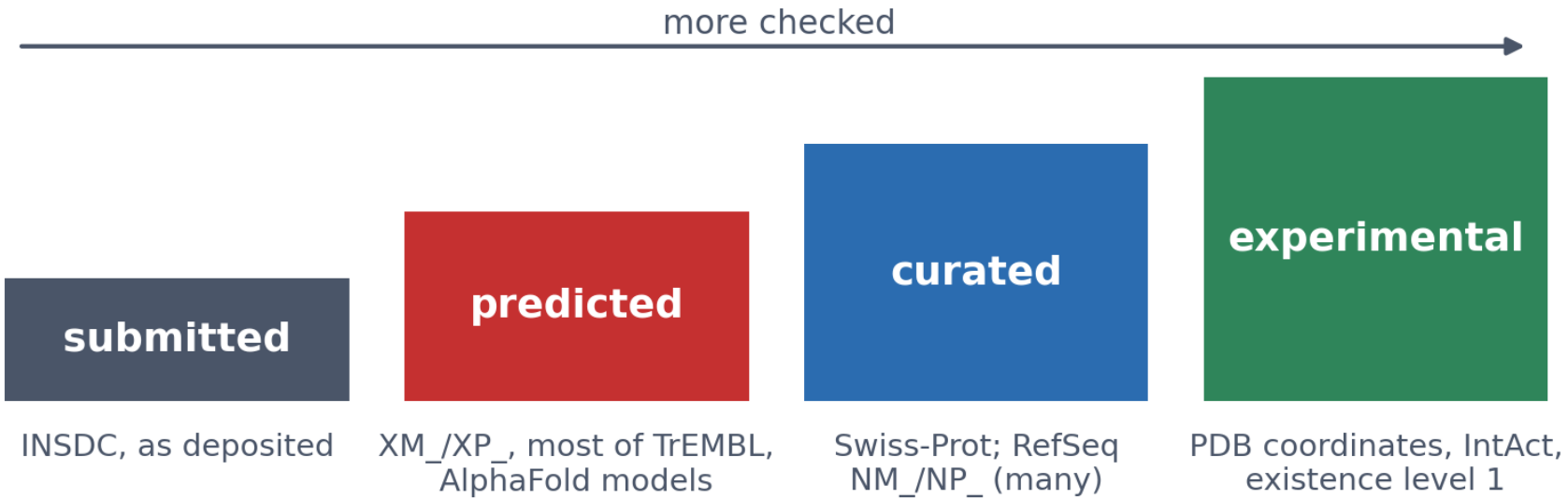
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BLOCK 2 · ABOUT 30 MIN

Who vouches for a record? HBB in five databases

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Submitted, predicted, curated, experimental

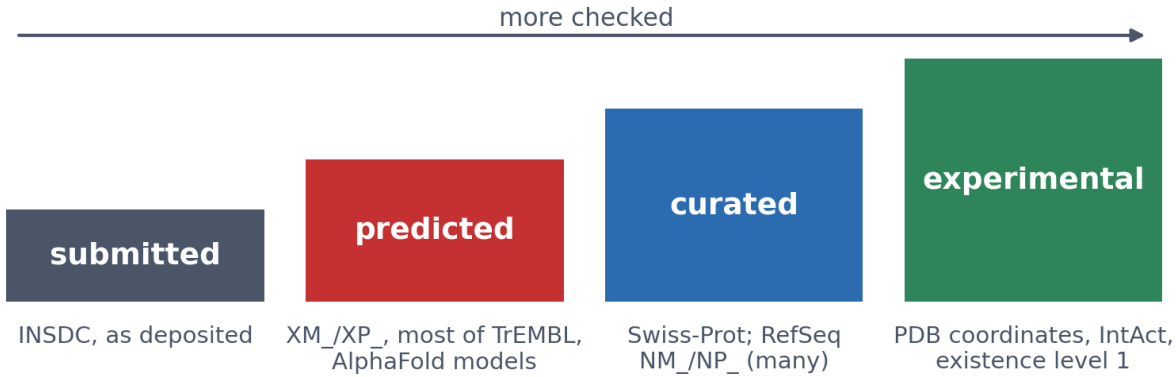


Sources of a claim, not kinds of record: one Swiss-Prot entry mixes curated, experimental and predicted annotation, and a PDB entry is both submitted and experimental.

See course.bioinfo.se/session03.html, 'Where does a record's information come from?'.

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Who vouches for each claim?



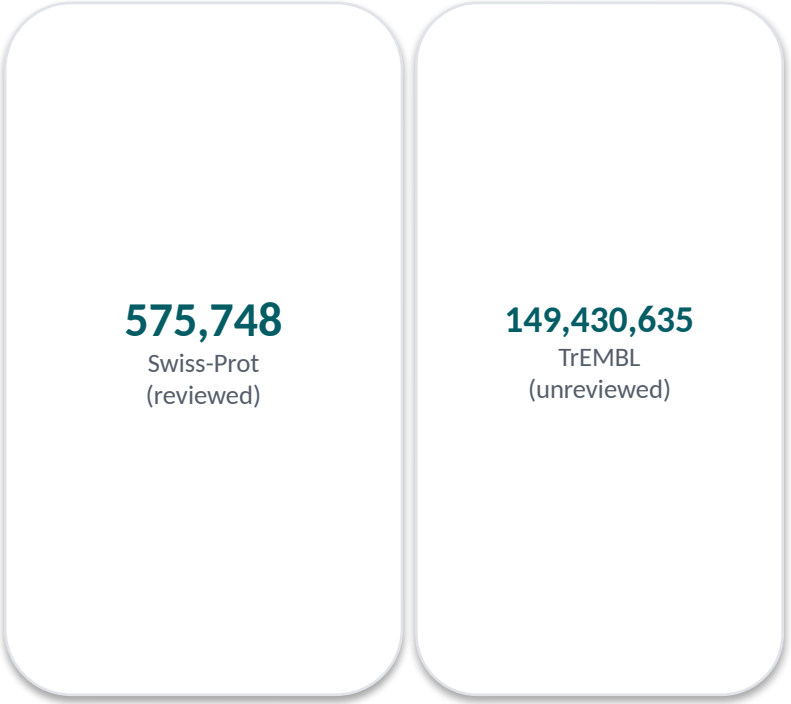
- The sequence of XP_011536997.1
- The atom positions in 2HHB
- The function text in P68871
- A sequence as deposited in ENA

See course.bioinfo.se/session03.html, 'Where does a record's information come from?'.

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Two sections: reviewed and unreviewed

- **Swiss-Prot: curators check function, domains, literature**
- TrEMBL: annotated automatically
- Not a subset: reviewed entries move to Swiss-Prot



Counts fetched live from the UniProt REST API; see the notebook.

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Protein existence: graded evidence

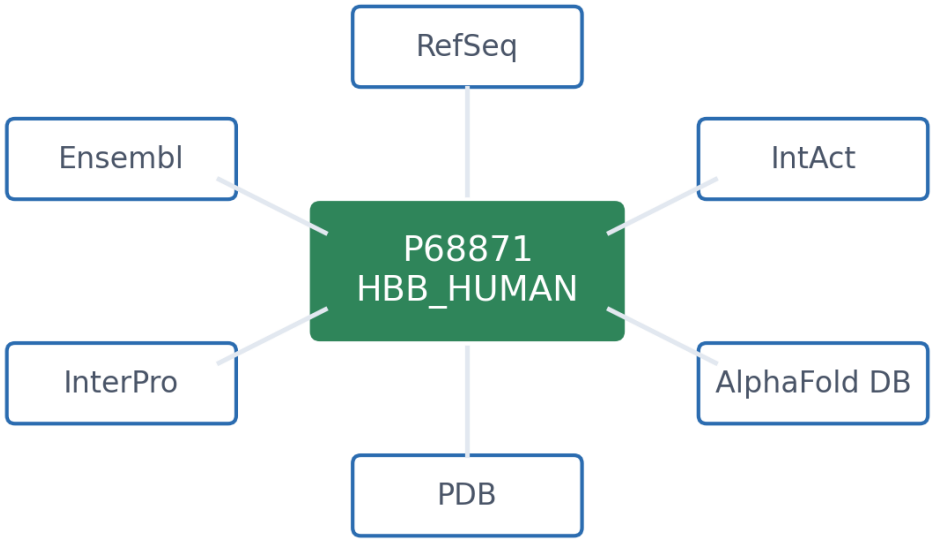
1. experimental evidence at protein level ← **HBB (P68871)**
2. experimental evidence at transcript level
3. inferred from homology
4. predicted
5. uncertain

- **Every entry has a level**
- From measured to uncertain
- HBB: level 1

UniProt; see course.bioinfo.se/session03.html.

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UniProt: the best first stop

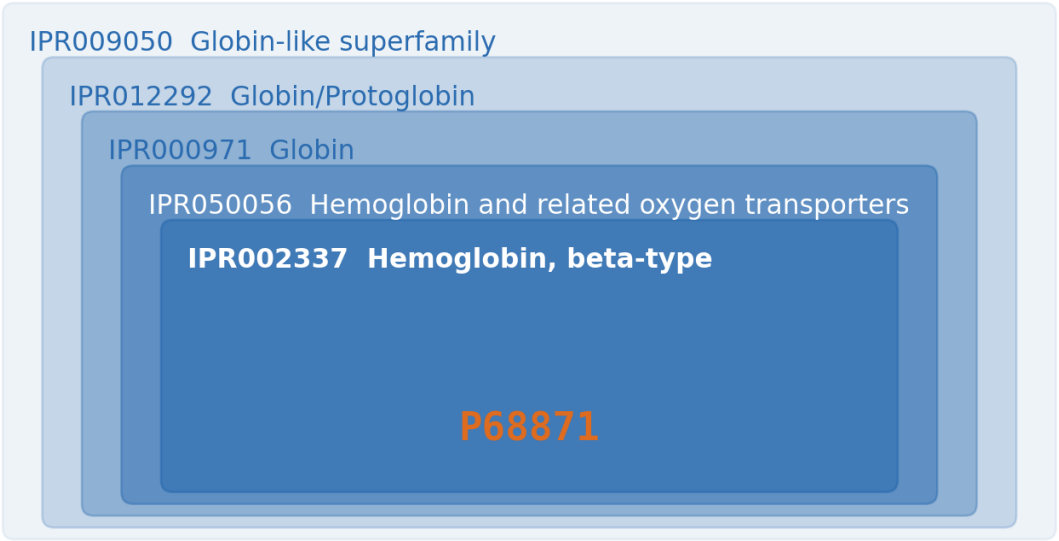


- **P68871: Swiss-Prot**
- Function, disease variants, modifications
- Links to everything else

See course.bioinfo.se/session03.html, 'What is known about the protein?'.

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InterPro: five nested entries for HBB

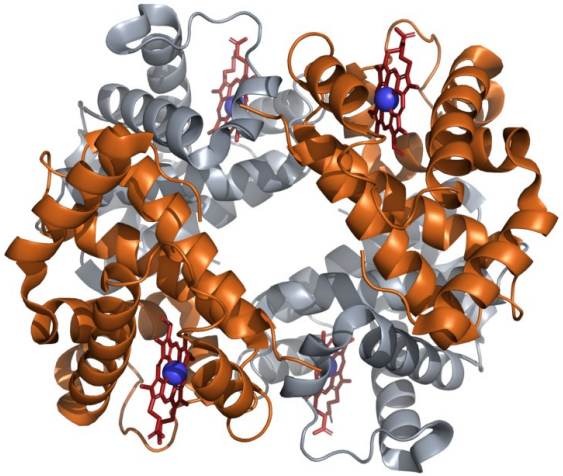


- **From superfamily to beta-type**
- Signatures from a dozen databases
- Pfam is now inside InterPro
- Mostly profile HMMs (Session 5)

InterPro API; see the notebook and book Table 3.3.

An experiment: PDB 2HHB

PDB 2HHB: X-ray, 1.74 Å (Fermi et al. 1984)

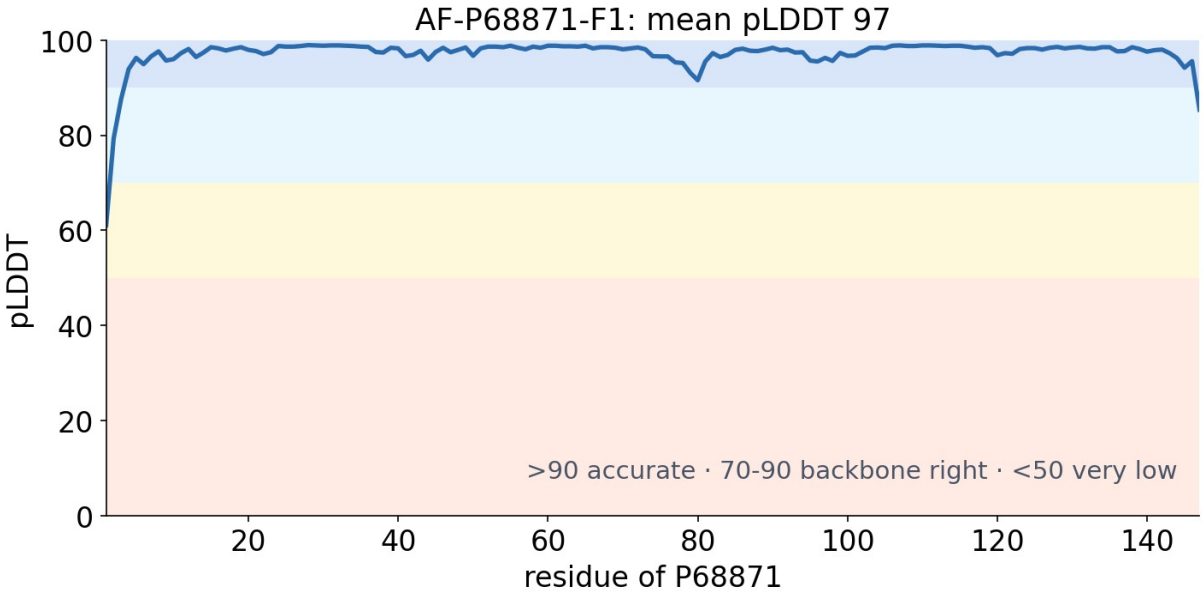


2 × HBA1 (α) 2 × HBB (β) 4 × haem

- **Human deoxyhaemoglobin**
- X-ray diffraction
- **Resolution 1.74 Å**
- Smaller = finer detail

PDBe and RCSB APIs; Fermi et al. (1984), J Mol Biol 175:159, PMID 6726807.

A prediction: AlphaFold DB



- **AF-P68871-F1**
- pLDDT: the model's own confidence
- Per residue, 0-100

AlphaFold Protein Structure Database (model v6); see the notebook.

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Resolution and pLDDT are different kinds of number

Resolution (PDB)

2HHB: 1.74 Å

- How much detail the experiment saw
- In a real molecule
- Smaller is better

pLDDT (AlphaFold DB)

AF-P68871-F1: mean 97

- The model's estimate of itself
- Per residue
- Larger is better

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What does the script print?

It searches UniProtKB for gene:HBB AND organism_id:9606 and averages the lengths. HBB has 147 residues.

```
hits = search_uniprot(
  "gene:HBB AND organism_id:9606")

lengths = [h["length"] for h in hits]
print(sum(lengths) / len(lengths))
```

Human haemoglobin beta (HBB) has 147 residues.

A 147: every hit is human HBB

B About 80: most hits are unreviewed fragments

C About 290: α and β counted together

D Nothing: one entry per gene

Guess on your own

Scan the code, or go to

quiz.bioinfo.se/live



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What does the script print?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

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What does the script print?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session03-discussion-2.ipynb

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What does the script print?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

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What does the script print?

Results, and who explains

Both rounds side by side.

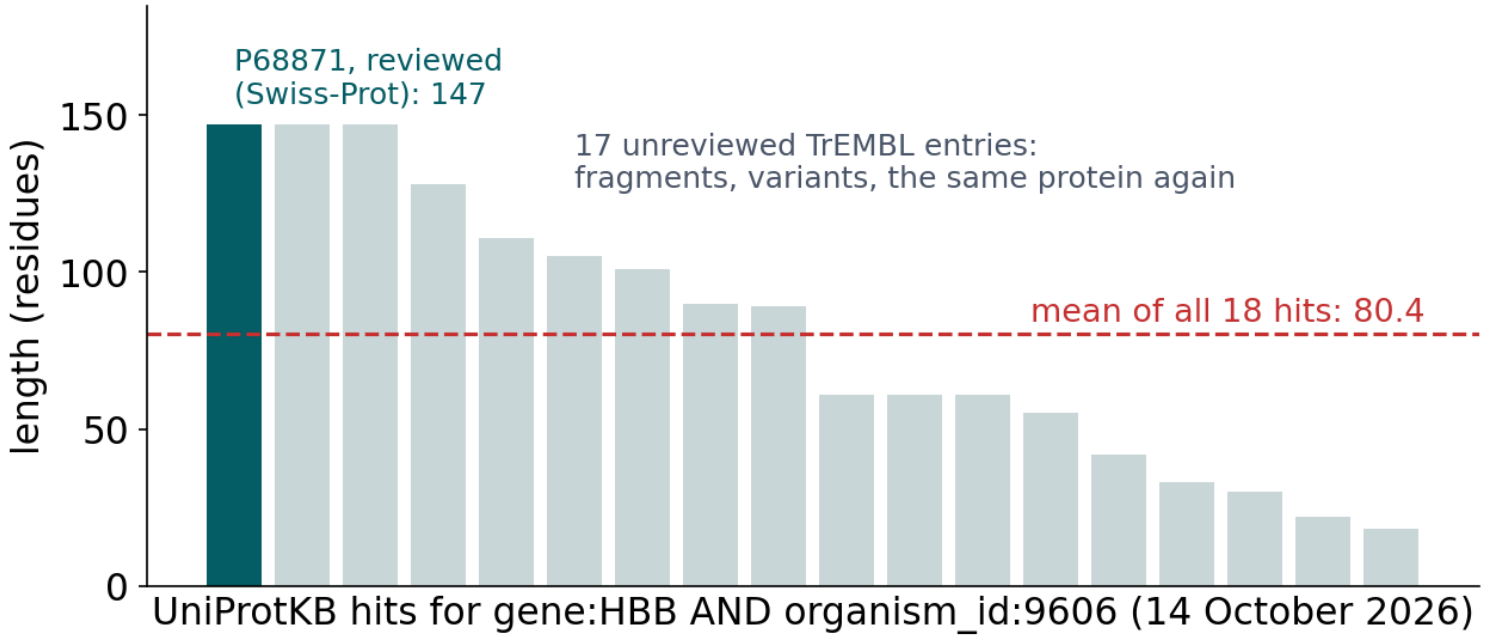
Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 2 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session03-discussion-2.ipynb

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The answer: about 80, because most hits are unreviewed



One reviewed entry (Swiss-Prot P68871, 147 residues) and 17 unreviewed TrEMBL entries, most of them fragments and variants from individual studies. A curator checks one reviewed entry per protein; unreviewed entries are added automatically, one per submitted sequence. Filter with reviewed:true, and look at the hits before you compute with them.

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BLOCK 2 DONE

Break

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BLOCK 3 · ABOUT 30 MIN

Interactions, the dossier, and searching

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Interactions: what kind of evidence?

IntAct	single physical interactions from published experiments	**HBB-HBZ**
Complex Portal	curated stable complexes with their composition	**CPX-2158 (HbA)**
STRING	one combined score: experiments, co-expression, predictions, text mining	**functional association**

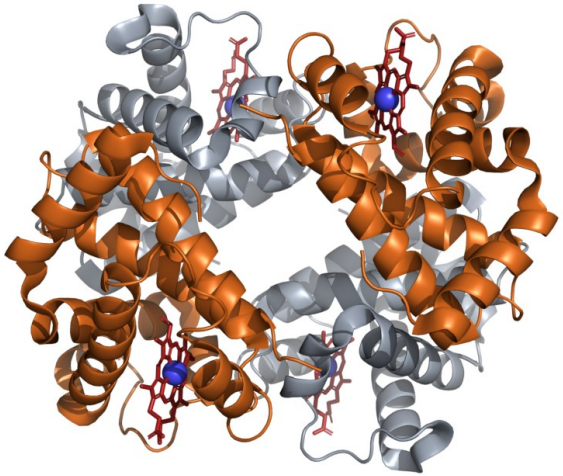
From the most to the least direct evidence. FunCoup (Stockholm University) builds a network like STRING's in a different way.

IntAct, Complex Portal and STRING APIs; see the notebook.

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A curated complex: CPX-2158, haemoglobin HbA

PDB 2HHB: X-ray, 1.74 Å (Fermi et al. 1984)



2 × HBA1 (α) 2 × HBB (β) 4 × haem

- **2 HBA1 + 2 HBB + 4 haem**
- Curated in Complex Portal
- The same object 2HHB shows

Complex Portal API; see the notebook.

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Five questions, five databases, one protein

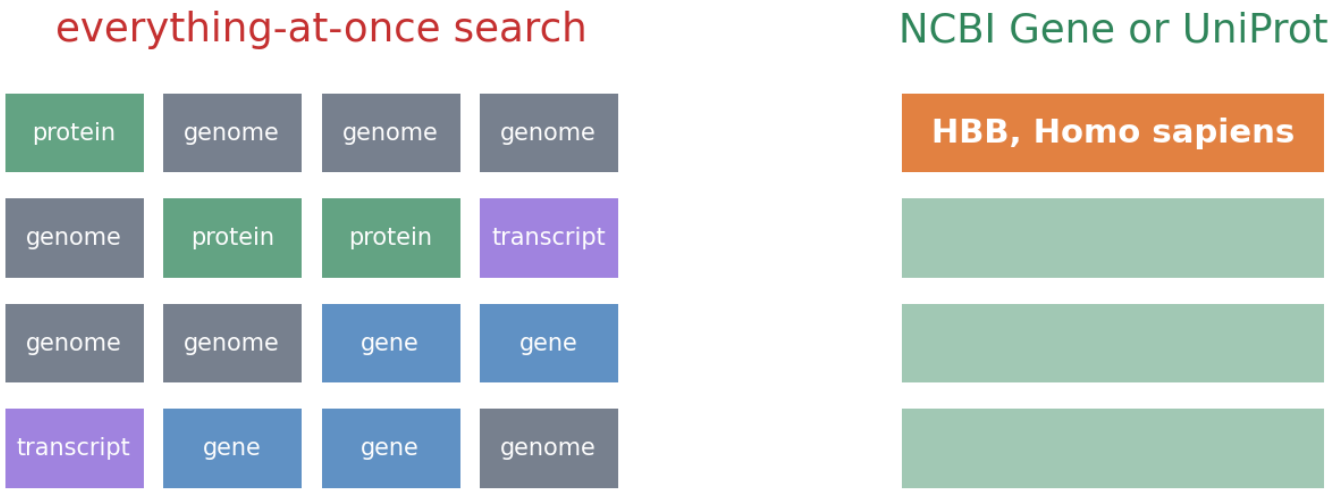
question	database	record	evidence
Where is the sequence?	RefSeq	NM_000518.5 → NP_000509.1	curated
What is known?	UniProt	P68871, existence level 1	curated
Which family?	InterPro	IPR000971 Globin (+4)	signatures
Is there a structure?	PDB · AFDB	2HHB 1.74 Å · AF-P68871-F1	exp. · predicted
What does it interact with?	IntAct · CPX	HBB-HBZ · CPX-2158	exp. · curated
Where is it described?	PubMed	PMID 6726807	literature

Each row is a different database's record of the same biology, reached from the previous one by a cross-reference. The notebook fetches every record live.

See course.bioinfo.se/session03.html, Table 3.4.

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Know your level, then search



An everything-at-once search mixes genome, gene, transcript and protein records. The same words in NCBI Gene or UniProt give a short list with the record you want near the top.

See course.bioinfo.se/session03.html, 'Searching'. The left panel is schematic.

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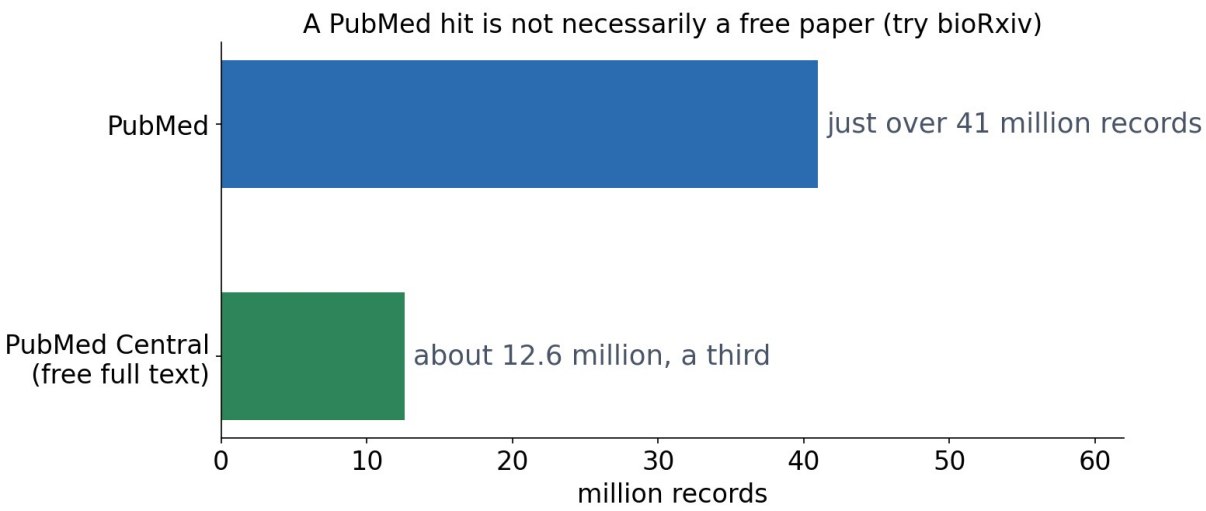
Searching from code: reproducible and scriptable

```
Entrez.email = 'you@student.su.se'
Entrez.efetch(db='protein', id='NP_000509.1', rettype='gb')
# → 147 residues, DBSOURCE: REFSEQ: accession NM_000518.5
```

```
requests.get('https://rest.uniprot.org/uniprotkb/P68871.json')
# → Swiss-Prot, HBB_HUMAN, protein existence 1
# reproducible: the same query gives the same records next year
```

NCBI's E-utilities through Bio.Entrez, and the UniProt and PDB REST APIs. The course notebook builds the whole HBB dossier this way.

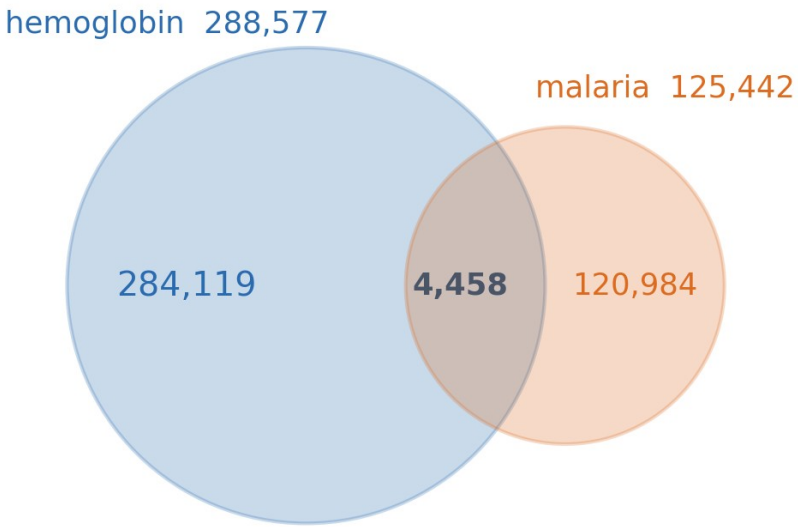
PubMed and PubMed Central



- **PubMed: built on MEDLINE (1971)**
- PMC: free full text
- No PMC copy? Try bioRxiv

SciLifeLab

Boolean operators are set operations



AND	4,458
OR	409,561
NOT	284,119

real PubMed counts

- **AND: the overlap**
- **OR: A + B − overlap**
- NOT: A − overlap

Live PubMed searches; see the notebook (book Table 3.5).

SciLifeLab

Quotation marks: words or a phrase?

```
hemoglobin beta
```

= hemoglobin AND beta: both words, anywhere, any order

```
"hemoglobin beta"
```

the exact phrase: usually far fewer records

- **Without quotes: AND**
- **With quotes: the phrase**
- Predict which gives fewer

See course.bioinfo.se/session03.html, 'Searching'.

SciLifeLab

Ask AI for leads, then go to the papers

- "How do intronic variants affect RNA splicing in human HBB?" — a mechanism to look for, and the words authors use for it
- "Find two primary studies that measured it" — now it has to name papers, not explain
- **"For each: the PMID, what was measured, and the figure that shows it" — an identifier to check, and a place to start reading**
- What the assistant actually had matters: learned text, a retrieved record, or a paper you supplied in the conversation

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Two checks, not one

Is this the paper?

- Title, authors, year match
- DOI or PMID resolves to that record
- **A working link proves nothing about the sentence beside it**

Is the claim supported?

- Find the result, not the abstract
- Which cells, which comparison, what was measured
- **The figure legend, Results and Methods belong together**

See course.bioinfo.se/session03.html, 'Put the claim beside the evidence'.

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Both checks have failed in print

- **A citation that does not exist**
  - Bhattacharyya and colleagues published ChatGPT output from April 2023: a 2021 sunscreen review by Chen et al., journal and all, carrying PMID 33635047
  - That PMID is real. It belongs to a paper on iron-oxide microspheres for enriching histidine-rich proteins
- **A real paper with the wrong conclusion**
  - Weinberg and colleagues asked for a summary of a named trial of saline and albumin in traumatic brain injury; the answer said there was no difference in mortality
  - The paper reports 33.2% against 20.4% at 24 months, P = 0.003 — the finding had been removed

Both cases are documented in the published literature; see the chapter for the links.

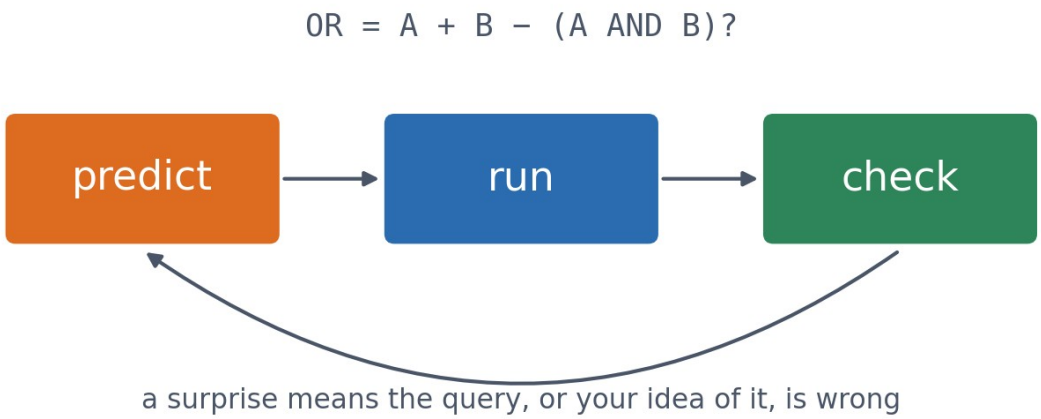
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Keep the conclusion inside the experiment

- Nualkaew et al. 2019, PMID 31113996: an engineered U7 snRNA against the IVS2-654 splice defect
- Cultured erythroid cells from patients, against untreated, vector and scrambled controls
- **RT-PCR, Figure 3A-B: a mean of 8.25% correctly spliced beta-globin RNA at day 10**
- That is RNA in that culture — not the share of cells corrected, and not a cure

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A habit for every database



- **Predict the count**
- Run the query
- Check that it makes sense

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Not needed in this course, but you will meet them

Variants

dbSNP, ClinVar, EVA

- GWAS Catalog
- OMIM for Mendelian disease

Cancer genomics

TCGA, COSMIC

- Via NCI's Genomic Data Commons
- Somatic mutations

Systems

PRIDE, Reactome, BioModels

- Proteomics data
- Pathways and models

See course.bioinfo.se/session03.html, 'Other resources worth knowing' (optional box).

SciLifeLab

Your own dossier: human myoglobin

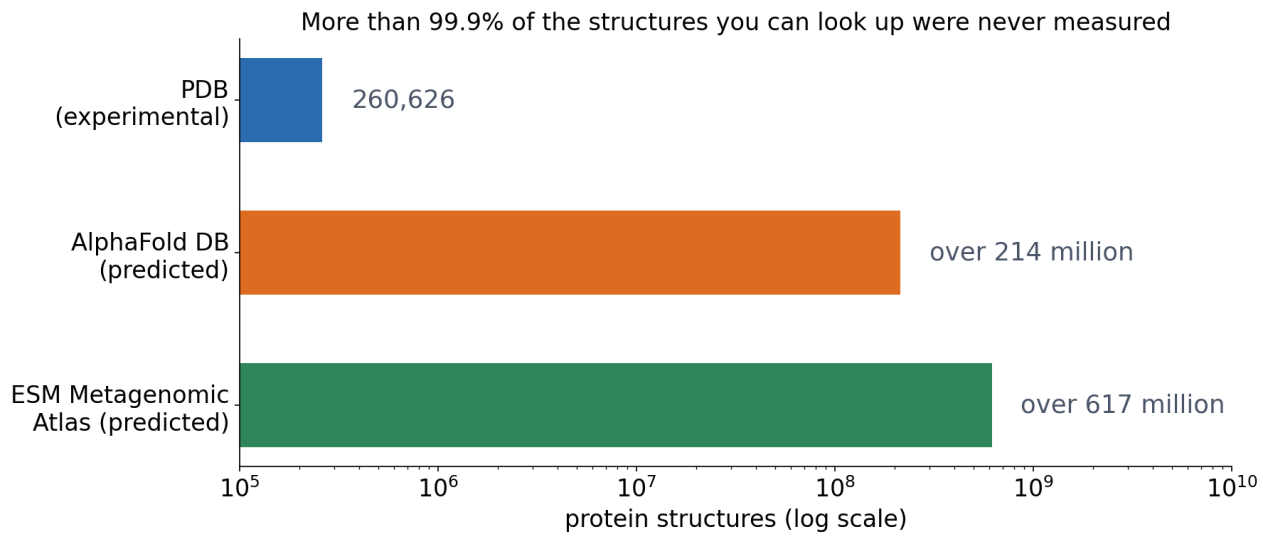
- identifiers and genome builds
- predicted and curated records
- UniProt entry, existence level
- InterPro domains
- structure and AlphaFold model
- complexes and networks
- a literature search

→ a dossier for human myoglobin

The same path on other genes and proteins, ending with a dossier in which you also say what kind of information each record holds.

The Session 3 lab quiz on Canvas (labs/session03-databases.text2qti.md).

From measured to predicted



- PDB: about 260,000 measured
- AlphaFold DB: over 214 million predicted
- ESM Atlas: over 617 million predicted
- **Who vouches for a predicted record?**

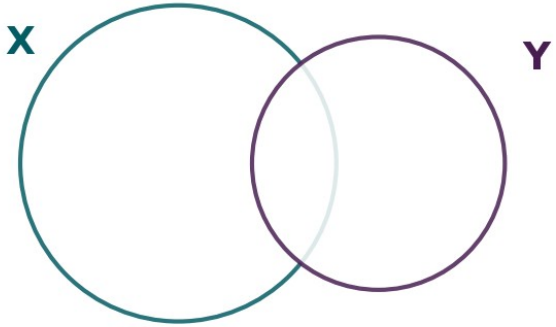
RCSB PDB search API, experimental entries, 9 Oct 2026; Varadi et al. 2024, NAR 52:D368; Lin et al. 2023, Science 379:1123. https://alphafold.ebi.ac.uk/

SciLifeLab

How many articles does Y NOT X give?

PubMed: X gives 50,000, Y 30,000, X AND Y 8,000, and X NOT Y 42,000.

X	50,000
Y	30,000
X AND Y	8,000
X NOT Y	42,000
Y NOT X	?



A 42,000: NOT works both ways alike

B 22,000

C 72,000

D 38,000

Guess on your own

Scan the code, or go to

quiz.bioinfo.se/live

SciLifeLab

How many articles does Y NOT X give?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

SciLifeLab

How many articles does Y NOT X give?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session03-discussion-3.ipynb

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How many articles does Y NOT X give?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

SciLifeLab

How many articles does Y NOT X give?

Results, and who explains

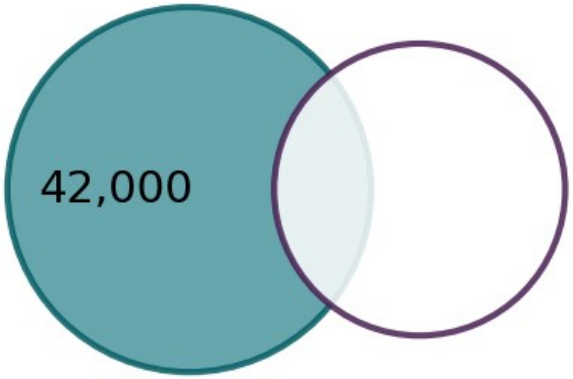
Both rounds side by side.

Draw two to explain: one of you says how you reasoned.

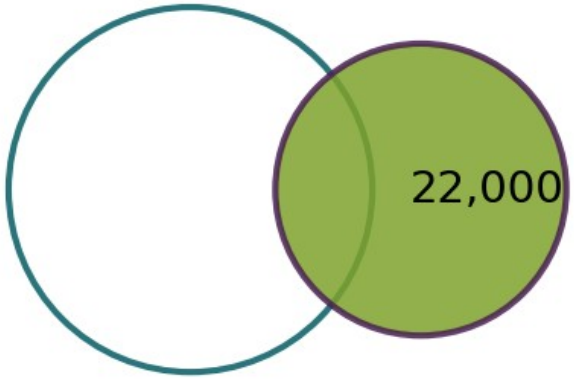
Show the results ▸ quiz.bioinfo.se/live/report

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The answer: 22,000, because NOT is not symmetric



X NOT Y = 50,000 − 8,000

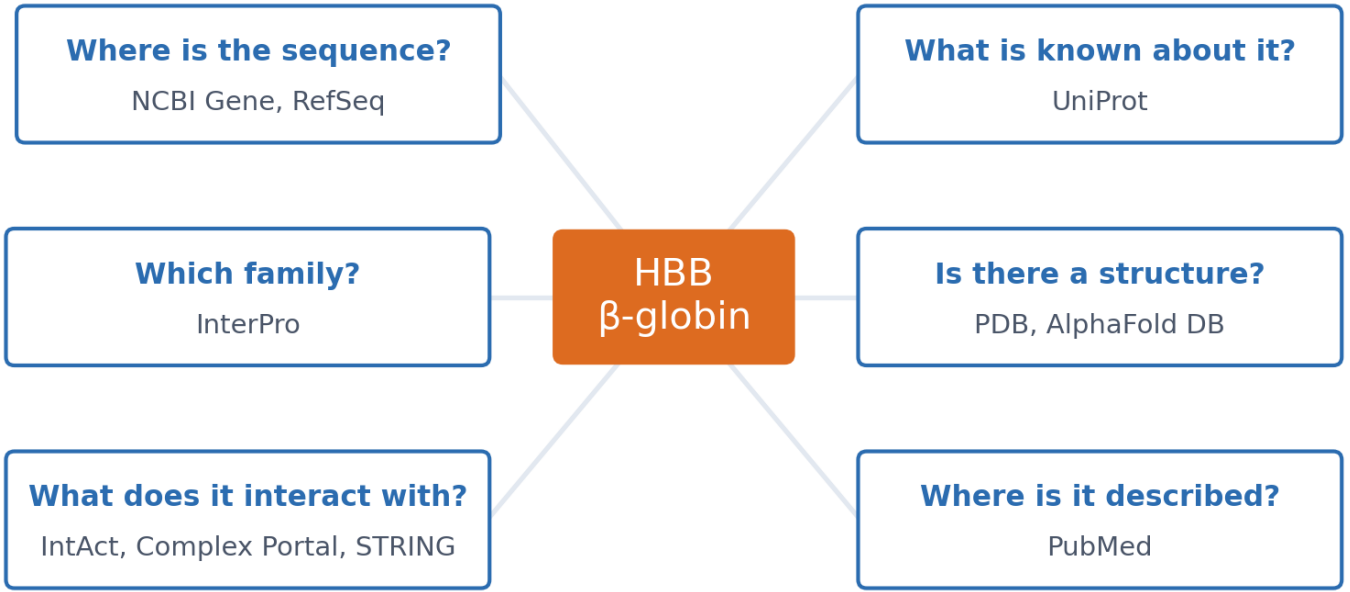


Y NOT X = 30,000 − 8,000

Boolean operators are set operations on the lists of matching articles. NOT starts from one set and removes the overlap, so X NOT Y = 50,000 − 8,000 and Y NOT X = 30,000 − 8,000. OR counts the overlap once: 72,000. Real PubMed counts follow the same arithmetic exactly.

SUMMARY

Know the identifier, know who vouches for it



Follow cross-references between records, report accession.version, ask where each claim comes from (submitted, predicted, curated, experimental), and predict a search before you run it.

See course.bioinfo.se/session03.html.

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In up to three words: what was the hardest part of today?

Identifiers

INSDC, RefSeq, accession.version

Who vouches?

UniProt, InterPro, PDB, AlphaFold DB

Searching

interactions, dossier, PubMed, Boolean

Answer now

Scan the code, or go to **quiz.bioinfo.se/live**

SciLifeLab

The hardest part of today

What the room found hardest

Show the results ▸ quiz.bioinfo.se/live/report

Session 4 next

You can now retrieve two related real sequences by accession
Next: how do you line them up, and score how similar they are?
Sequence alignment: dynamic programming, PAM/BLOSUM, dot plots